GOLD PRICE FORECASTING REPORT

GLOBAL SUPPLY CHAIN COMMODITY PRICE FORECASTING & VOLATILITY ANALYTICS SYSTEM

MACHINE LEARNING & TABLEAU PROJECT

PROJECT REPORT

BY

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TECHOLAS TECHNOLOGIES

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1. INTRODUCTION

Gold has long been regarded as a safe-haven asset and a barometer of global economic uncertainty. Its price is influenced by a complex interplay of macroeconomic factors including inflation, currency movements, geopolitical events, and supply chain dynamics. Accurately forecasting gold prices offers significant value to investors, commodity traders, central banks, and supply chain analysts who rely on price signals for procurement and hedging decisions. This project develops a Global Supply Chain Commodity Price Forecasting and Volatility Analytics System that applies machine learning regression techniques to historical gold OHLCV (Open, High, Low, Close, Volume) data. The system performs exploratory analysis, volatility engineering, multi-model comparison, and deploys the best model as an interactive Streamlit forecasting tool.

Why Gold Price Forecasting Matters

For Investors	Informs portfolio allocation and hedging strategies
For Supply Chains	Enables better procurement planning for gold-intensive industries
For Central Banks	Supports reserve management and monetary policy decisions
For Data Science	Demonstrates time-series regression with real financial data

2. ABSTRACT

This project develops a machine learning-based gold price forecasting system using historical OHLCV data. After performing Exploratory Data Analysis (EDA), volatility analytics, and feature engineering (Year, Month, Day extraction), four regression algorithms—Linear Regression, Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor—were evaluated using GridSearchCV with R^2 as the scoring metric. Linear Regression achieved the highest cross-validation R^2 score of 0.9999, outperforming all ensemble models on this dataset, and was selected as the final forecasting model. The trained model was saved with joblib and deployed as a Streamlit web application for interactive gold price prediction.

3.OBJECTIVES

- **Analyse Gold Price Data:**
  Explore historical OHLCV data including trends, price distributions, and temporal patterns.

- **Perform Volatility Analytics:**
  Engineer daily price change and volatility features; identify the most volatile trading day

- **EDA & Feature Engineering:**
  Derive Year, Month, Day features; analyse correlations between OHLCV variables.

- **Build Regression Models :**
  Train and tune Linear Regression, Decision Tree, Random Forest, and Gradient Boosting regressors.

- **Select & Deploy Best Model:**
  Choose Linear Regression (highest R^2) and deploy it as a Streamlit forecasting app.

4. METHODOLOGY

Step 1 – Data Collection:

Historical gold price dataset (gold.csv) containing Date, Open, High, Low, Close, and Volume fields.

Step 2 – Data Pre-processing :

Converted Date column to datetime format. Confirmed zero missing values. Verified data types. Extracted start and end dates to understand temporal coverage.

Step 3 – Exploratory Data Analysis :

Plotted closing price trend over time (line chart). Scatter plot of Open vs Close. Distribution of Close prices (histplot with KDE). OHLCV correlation heatmap. Box plot of Close prices for outlier identification.

Step 4 – Volatility Analytics Engineered :

Price_Change = Close − Open (daily direction indicator). Engineered Volatility = High − Low (intraday range). Identified Top 10 most volatile days. Calculated average daily volatility (~16.48 units)

Step 5 – Feature Engineering Extracted :

Year, Month, Day from the Date column. Final feature set: Open, High, Low, Volume, Year, Month, Day (7 features). Target variable: Close price.

Step 6 – Model Building & Hyperparameter Tuning :

Applied GridSearchCV (3-fold CV, scoring="r2") across four regressors: Linear Regression (no hyperparameters), Decision Tree (max_depth ∈ {5,10,15}, min_samples_split ∈ {2,5,10}), Random Forest (n_estimators ∈ {50,100}, max_depth ∈ {5,10}), Gradient Boosting (n_estimators ∈ {50,100}, learning_rate ∈ {0.05,0.1}, max_depth ∈ {3,5}).

Step 7 – Model Saving & Deployment:

Best model (Linear Regression) persisted as gold_price_forecasting_model.pkl using joblib. Deployed as a Streamlit web application for interactive price forecasting

5. DATASET OVERVIEW

- ★ **Source:** Kaggle
- ★ **Features:** 7 (inputs)
- ★ **Target:** Close Price
- ★ **Task Type:** Regressio n

The dataset contains historical daily gold price records with standard OHLCV structure. Date spans from early 2000s to recent years, providing a rich longitudinal dataset covering multiple economic cycles including the 2008 financial crisis, 2013 gold crash, and post-COVID rally.

6. DATA ANALYSIS & EDA

Long-term Trend

Gold closing prices show a sustained upward trend over the observation period, with notable spikes during periods of economic uncertainty.

Open vs Close

Strong linear relationship between Open and Close prices — a scatter plot reveals near-perfect correlation, consistent with market continuity.

Price Distribution

The histogram (with KDE) shows a right-skewed distribution of Close prices, indicating that extreme high prices are relatively rare but exert strong influence.

OHLCV Correlations

Open, High, Low, and Close are all very highly correlated (>0.99) with each other, as expected for financial price data. Volume shows a weak negative correlation with price level.

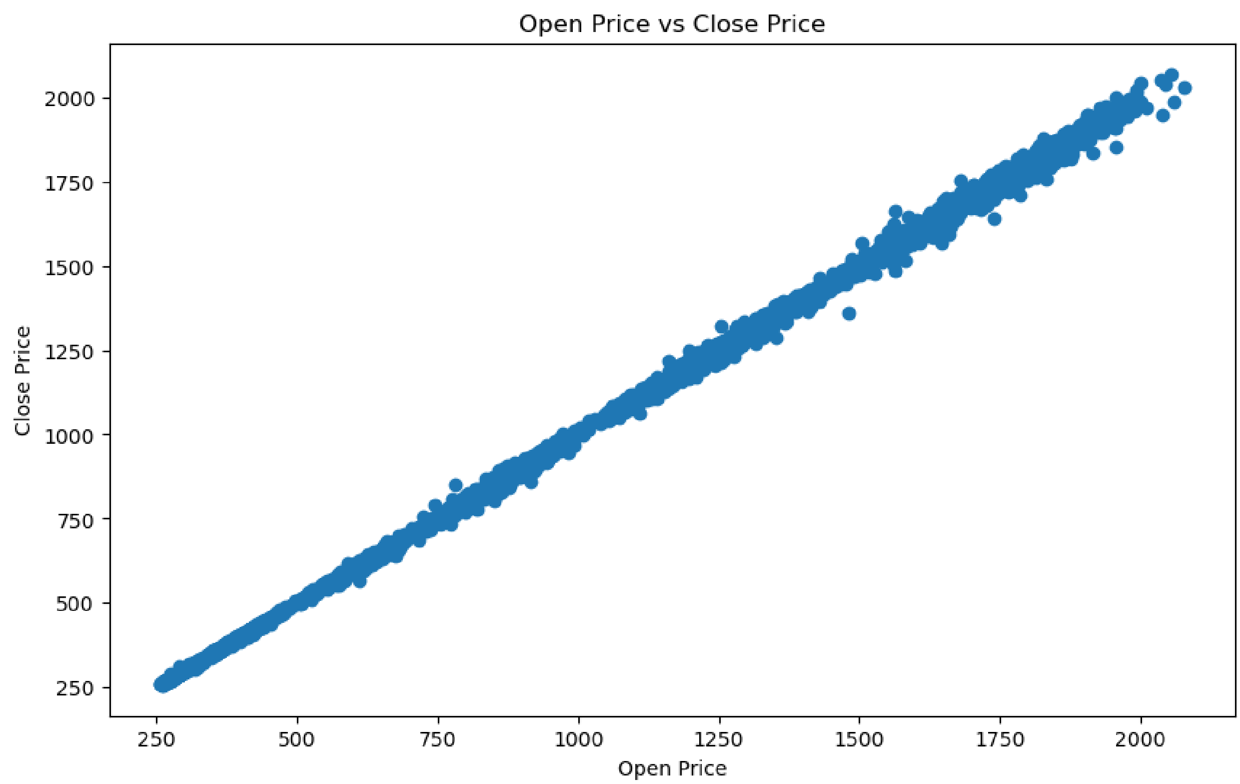
Outlier Analysis

Box plot of Close reveals a long upper whisker but no strict outliers by IQR convention — price extremes reflect genuine market events rather than data errors.

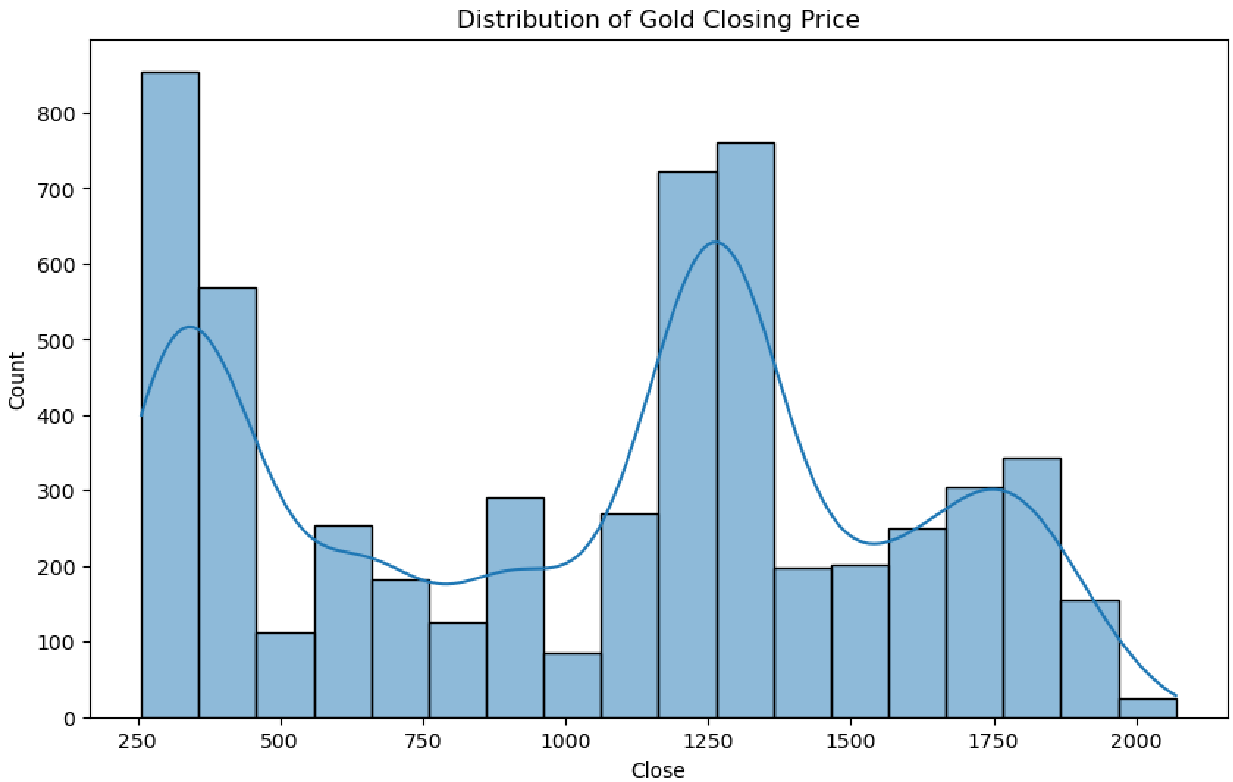
Visualisations Used

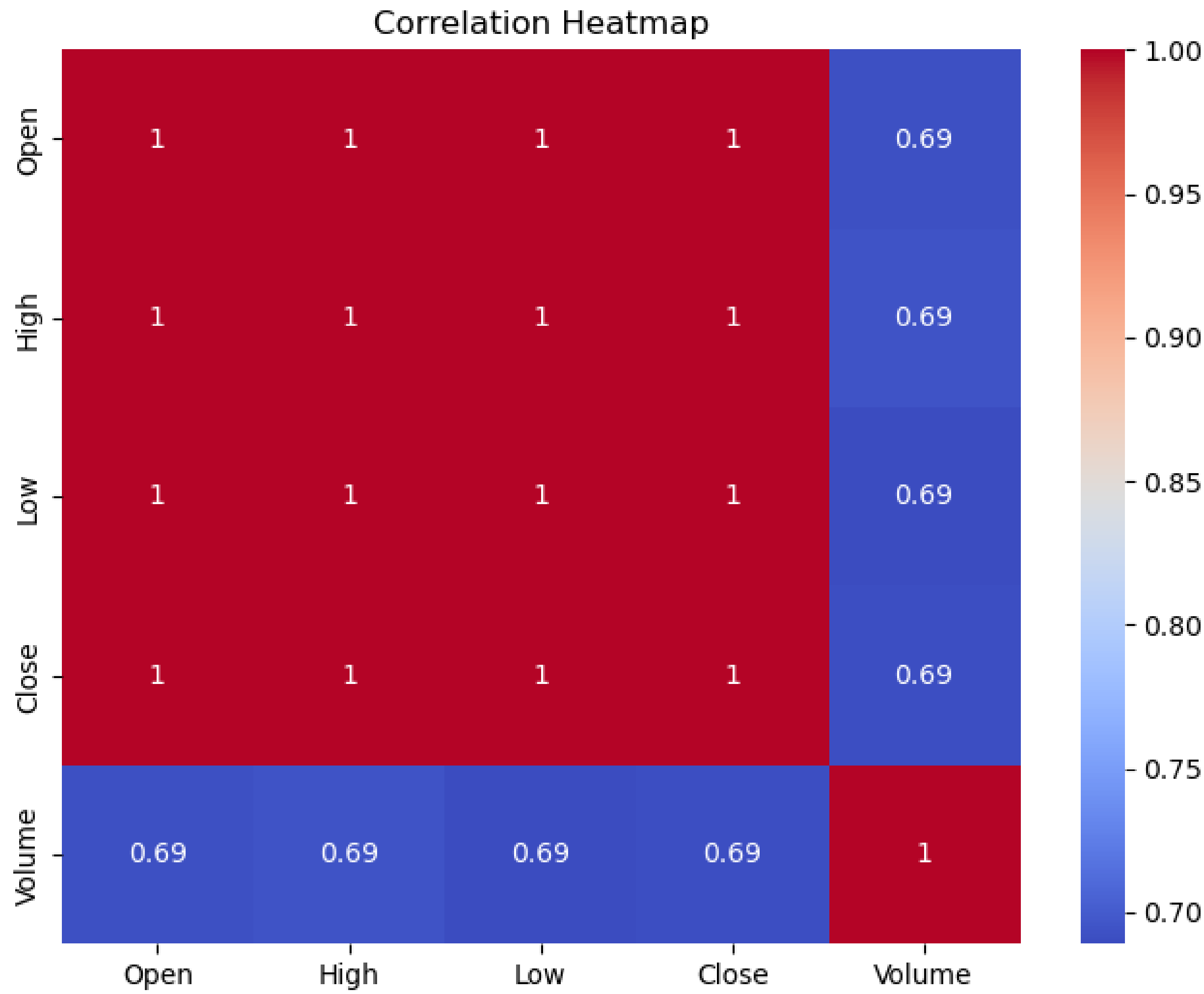
- Line Chart — Gold closing price trend over time
- Scatter Plot — Open vs Close price relationship
- Histogram (KDE) — Distribution of gold closing prices
- Correlation Heatmap — OHLCV inter-feature relationships
- Box Plot — Closing price outlier detection

Gold Closing Price Trend
Close Price
Date
250
500
750
1000
1250
1500
1750
2000
2000
2004
2008
2012
2016
2020



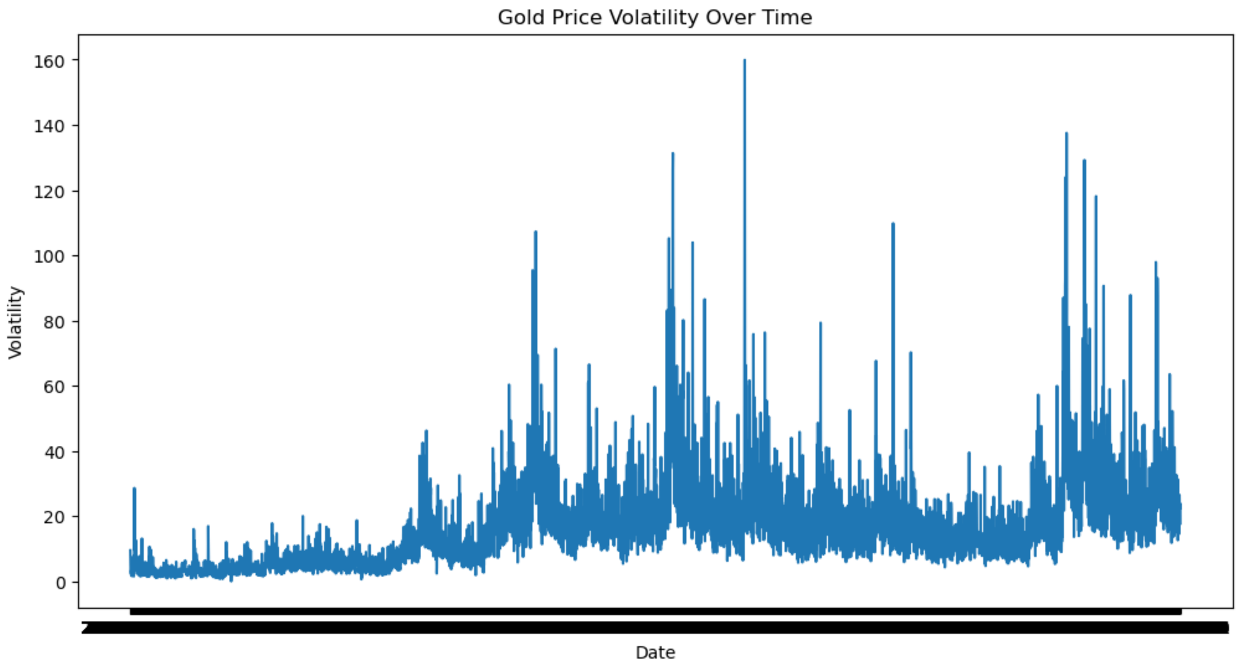
Open Price vs Close Price
Close Price
Open Price
250
500
750
1000
1250
1500
1750
2000
250
500
750
1000
1250
1500
1750
2000


Distribution of Gold Closing Price
Count
Close
0
100
200
300
400
500
600
700
800
250
500
750
1000
1250
1500
1750
2000


Correlation Heatmap
Open
High
Low
Close
Volume
1
0.69
1.00
0.95
0.90
0.85
0.80
0.75
0.70

7. VOLATILITY ANALYTICS

Volatility analysis is a key component of commodity price intelligence. Two engineered features were derived to quantify daily price behaviour:



Key Volatility Findings

Volatility Trend:

The volatility generally shows an increasing trend over time. This suggests that gold price fluctuations became larger during later years compared to earlier years.

Most Volatile Day:

5 April 2013 recorded the highest single-day volatility at 159.9 units — corresponding to the historic gold price crash triggered by fears of Cyprus selling gold reserves.

Average Volatility:

The mean intraday volatility across the full dataset was ~16.48 units, providing a baseline for identifying abnormally volatile trading sessions.

8. MODEL DEVELOPMENT & RESULTS

Models Applied:

1. Linear Regression
2. Random Forest
3. Gradient Boosting
4. Decision Tree

GridSearchCV Model Comparison (3-Fold, R^2 Scoring)

	model	best_score	best_params
0	linear_regression	0.999918	{}
2	random_forest	0.999837	{'max_depth': 10, 'n_estimators': 100}
3	gradient_boosting	0.999775	{'learning_rate': 0.1, 'n_estimators': 100}
1	decision_tree	0.999727	{'max_depth': 10, 'min_samples_split': 10}

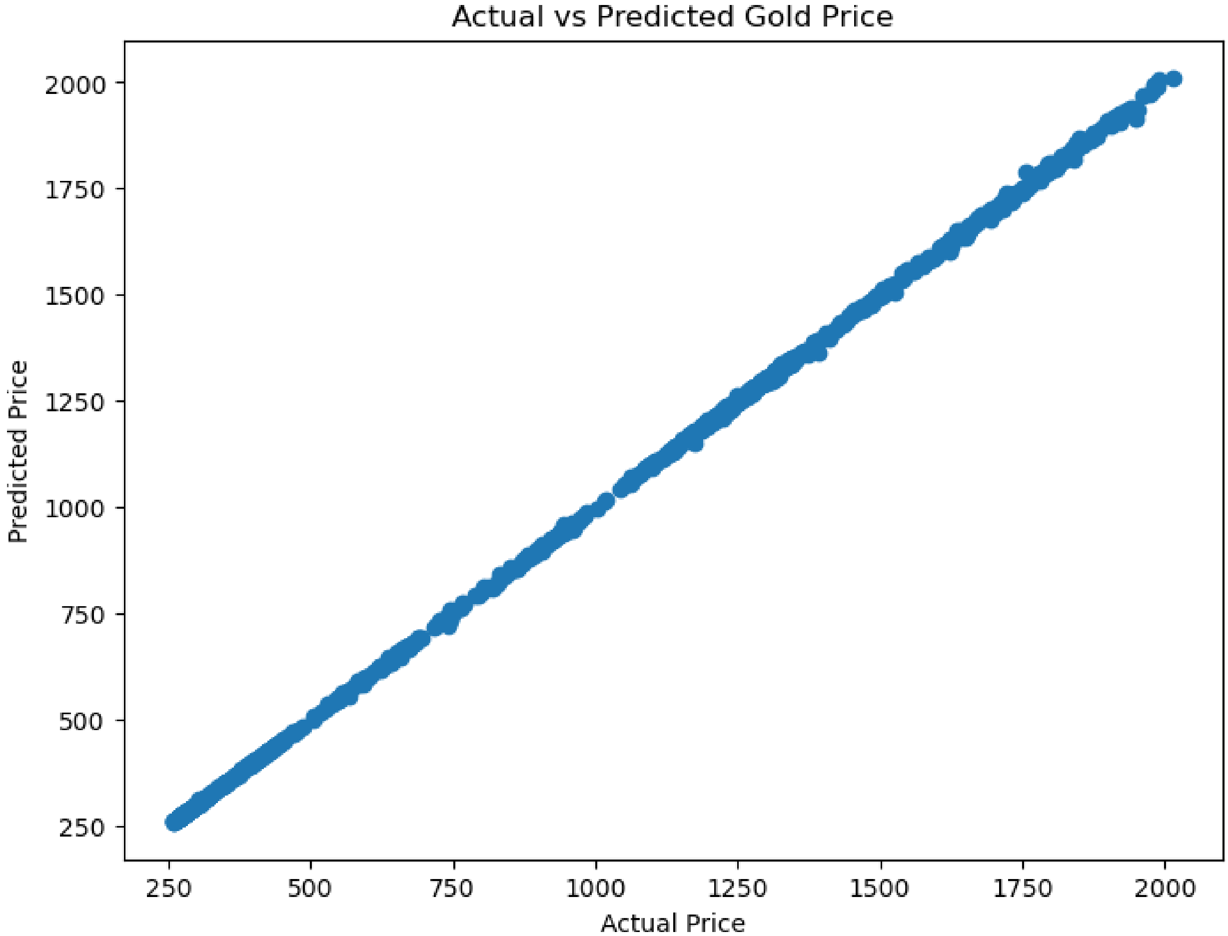
Final Model — Linear Regression (Performance on Test Set)

```
MAE : 2.9824836159369084
MSE : 21.964121437183785
RMSE : 4.686589531544637
R2 Score : 0.9999219031380453
```

The Linear Regression model performed exceptionally well on the Gold Price dataset. The model achieved an R^2 Score of 0.99992, indicating that it explained more than 99.99% of the variance in gold prices. The low MAE (2.98) and RMSE (4.69) values suggest highly accurate predictions.

Actual vs Predicted Analysis

The scatter plot of Actual vs Predicted Close prices shows near-perfect alignment along the diagonal, confirming that the model generalises well without overfitting. No systematic bias or heteroscedasticity was observed across different price levels.



9.DASHBOARD OVERVIEW

The **Global Supply Chain Commodity Price Forecasting and Volatility Analytics Dashboard** is an interactive business intelligence dashboard developed to analyze historical gold price trends, measure market volatility, and support commodity price forecasting. The dashboard provides key insights into gold price movements over time, helping analysts, investors, and decision-makers understand market behavior and identify trends that may impact supply chain and investment decisions.

KPI

The dashboard presents important metrics that summarize the overall performance of gold prices:

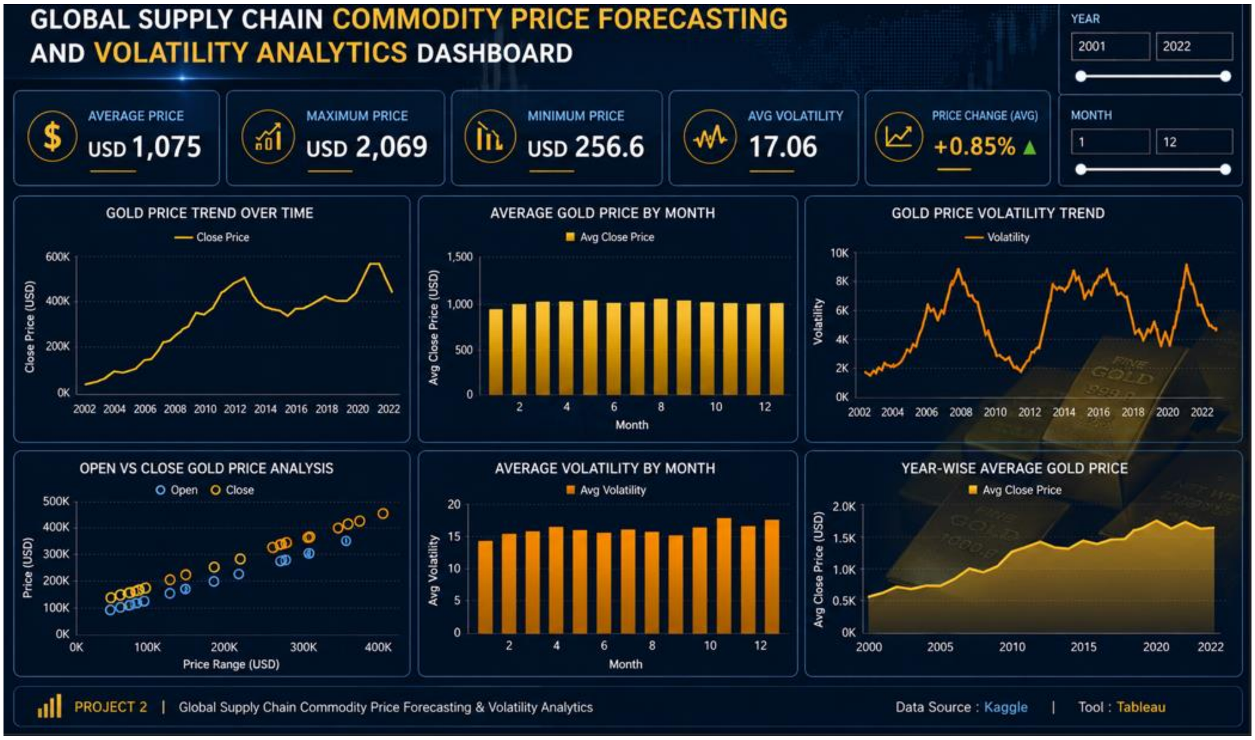
- **Average Price:** USD 1,075
- **Maximum Price:** USD 2,069
- **Minimum Price:** USD 256.6
- **Average Volatility:** 17.06
- **Average Price Change:** +0.85%

Visualizations

1.Gold Price Trend Over Time

- Displays the historical movement of gold prices from 2001 to 2022.
- Helps identify long-term growth patterns, market fluctuations, and major price shifts.

2. Average Gold Price by Month
    - Shows the average gold price for each month.
    - Enables comparison of monthly price patterns and seasonal trends.
3. Gold Price Volatility Trend
    - Illustrates changes in market volatility over time.
    - Helps identify periods of high uncertainty and significant market movements.
4. Open vs Close Gold Price Analysis
    - Compares opening and closing prices using a scatter plot.
    - Assists in understanding daily market behavior and price variations.
5. Average Volatility by Month
    - Displays monthly average volatility levels.
    - Highlights months that experience relatively higher or lower market fluctuations.
6. Year-wise Average Gold Price
    - Shows the average gold price for each year.
    - Provides a clear view of long-term price growth and yearly market performance.


GLOBAL SUPPLY CHAIN COMMODITY PRICE FORECASTING AND VOLATILITY ANALYTICS DASHBOARD
YEAR
2001
2022
MONTH
1
12
AVERAGE PRICE
USD 1,075
MAXIMUM PRICE
USD 2,069
MINIMUM PRICE
USD 256.6
AVG VOLATILITY
17.06
PRICE CHANGE (AVG)
+0.85%
GOLD PRICE TREND OVER TIME
Close Price
Close Price (USD)
AVERAGE GOLD PRICE BY MONTH
Avg Close Price
Avg Close Price (USD)
Month
GOLD PRICE VOLATILITY TREND
Volatility
OPEN VS CLOSE GOLD PRICE ANALYSIS
Open
Close
Price (USD)
Price Range (USD)
AVERAGE VOLATILITY BY MONTH
Avg Volatility
Month
YEAR-WISE AVERAGE GOLD PRICE
Avg Close Price
Avg Close Price (USD)
PROJECT 2 | Global Supply Chain Commodity Price Forecasting & Volatility Analytics
Data Source : Kaggle | Tool : Tableau

10.STREAMLIT WEB APP DEPLOYMENT

The final Linear Regression model was persisted using joblib as gold_price_forecasting_model.pkl and integrated into a Streamlit web application for interactive gold price forecasting.

📈 Global Supply Chain Commodity Price Forecasting

Gold Price Prediction System

Enter Commodity Details

Open Price

2000.00 − +

High Price

2015.00 − +

Low Price

1990.00 − +

Volume

10000.00 − +

Year

2024 − +

Month

12 − +

Day

25 − +

Predict Gold Price

💰 Predicted Gold Close Price: 2005.56

CONCLUSION & FUTURE SCOPE

Conclusion

This project successfully developed a Global Supply Chain Commodity Price Forecasting and Volatility Analytics System. The full data science pipeline was executed: EDA, volatility feature engineering, multi-model regression comparison using GridSearchCV, and model deployment. Among all models tested, Linear Regression achieved the highest R^2 of 0.99992 and was selected as the final forecasting model. The volatility analytics component additionally identified the most turbulent trading days and characterised average intraday price range, offering actionable intelligence for commodity risk managers and supply chain planners

Future Scope

• Incorporate macroeconomic indicators (USD index, CPI, oil prices) as additional input features.

• Apply time-series models (ARIMA, LSTM) to capture autocorrelation and sequential dependencies in gold prices.

• Develop a real-time data pipeline using financial APIs (Yahoo Finance, Alpha Vantage) for live forecasting.

• Extend the system to forecast other commodities (silver, platinum, crude oil) within a unified dashboard.

• Add SHAP-based feature attribution to explain individual price predictions to non-technical stakeholders.

• Integrate ensemble stacking (meta-learner combining multiple base models) for potentially higher accuracy